

ENERGY 295: Final Project

SPM Discretization, Analysis and State Estimation

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Oral Presentation: March 11, 2026 (in class)

Final Report: March 16, 2026 by 5pm (on Canvas)

Overview

This final project builds upon the Midterm Project (FDM implementation of the SPM) and gives you the opportunity to explore advanced topics in battery modeling and state estimation. You will choose a project direction that interests your group, develop a rigorous analysis, and present your findings.

Students must work within their original midterm groups. This project builds upon the provided SPM code located in the `SPM/` directory available on Canvas.

Provided Resources

- Working SPM implementation with CasADi (`SPM/cell_model/`) with a README file for context (completed in the midterm project)
- Parameters for LG Chem INR21700 cells (`modelParameters.m`) [\[1\]](#)
- Experimental data: HPPC and UDDS drive cycles (`SPM/data/`) [\[2\]](#)
- Helper functions for concentrations, OCV curves, visualization (`SPM/cell_model/`)
- Empty template: `fvm_matrices.m` (available for FVM-related projects)
- Simulation scripts: `run_sim_MATLAB.m` and `run_sim_CasADi.m`

Final Project Descriptions

Goal

Choose a project direction that explores numerical methods for battery modeling, state estimation techniques, or a combination of both. Your project should demonstrate technical depth, rigorous analysis, and clear communication of results. The following are only high-level directions. You are encouraged to define your own specific research questions and methodology within these broad themes, and to check in with Prof. Onori and the Sai for feedback and guidance as you develop the project.

Project Directions

Below are several example project directions. You may choose one of these, combine elements from multiple options, or propose your own project (subject to instructor approval). Each example includes suggested scope and analysis directions, but you are encouraged to define your own specific research questions and methodology.

Option 1: FDM vs. FVM Performance Comparison

Implement the Finite Volume Method (FVM) (see the following background section for details) for the SPM and conduct a rigorous comparison with your FDM implementation from the Midterm Project.

Possible Analysis Directions:

- Grid convergence study: accuracy vs. number of nodes for both methods
- Computational efficiency: timing comparisons under various conditions
- Conservation properties: verify lithium mass conservation
- Observability analysis and condition number comparisons between methods (see Lucero & Onori, ACC 2025 [4])
- Recommendations for when to use each method in practice

Option 2: Extended Kalman Filter (EKF) Implementation for SOC Estimation

Develop an Extended Kalman filter (EKF) for real-time state-of-charge estimation using the SPM as the underlying model [6].

Possible Analysis Directions:

- Implement EKF or AEKF with the reduced order SPM (i.e. SPM with fewer nodes). Show how you select the nodes (states) for the reduced order model.
- Analyze performance on different drive cycles (UDDS, HPPC) against the full SPM solution (101 nodes) and experimental data as ground truth
- Study the effect of different initial conditions on convergence and accuracy
- Tune the Q and R matrices and clearly show the steps taken to calibrate them

Option 3: Comparison of Kalman Filter Techniques

Implement at least 2 Kalman filter variants and compare their performance for battery state estimation.

Filters to Compare:

- **Extended Kalman Filter (EKF):** Linearizes using Jacobians; fast but requires derivative computation
- **Adaptive EKF:** Dynamically adjusts $\mathbf{Q}$ and R based on innovation statistics
- **Unscented Kalman Filter (UKF):** Uses sigma points; handles strong nonlinearities better

Possible Analysis Directions:

- Implement 2 filter variants with consistent 4-node SPM model
- Compare SOC estimation accuracy (RMSE, max error) with the SoC computed from the experimental data as ground truth
- Analyze convergence speed from large initial errors under the UDDS drive cycle
- Evaluate computational cost per time step
- Identify scenarios where each filter excels or struggles

Option 4: Propose Your Own Project

If you have a different idea, propose your own project direction related to SPM modeling, numerical methods, or state estimation. Discuss with the instructor/TA by Friday, 13th February 2026 for approval.

Background: Finite Volume Method

FDM approximates derivatives at spatially discretized points. FVM takes a different approach—it integrates the governing equation over control volumes to ensure conservation. For each electrode $j \in \{n, p\}$, we integrate the spherical diffusion equation over shells:

$$\int_{V_j} \frac{\partial c_{s,j}}{\partial t} dV = \int_{V_j} \frac{D_{s,j}}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_{s,j}}{\partial r} \right) dV, \quad j \in \{n, p\}$$

Applying the divergence theorem and assuming cell-averaged concentrations $\bar{c}_{s,j,i}$ in i -th shell gives (see [3] for details):

$$V_{j,i} \frac{d\bar{c}_{s,j,i}}{dt} = A_{j,i+1/2} J_{j,i+1/2} - A_{j,i-1/2} J_{j,i-1/2}, \quad j \in \{n, p\}$$

where:

- $V_{j,i} = \frac{4\pi}{3}(r_{j,i+1/2}^3 - r_{j,i-1/2}^3)$ is the shell volume in electrode j
- $A_{j,i+1/2} = 4\pi r_{j,i+1/2}^2$ is the interface area in electrode j
- $J_{j,i+1/2} = -D_{s,j} \frac{\partial c_{s,j}}{\partial r} \Big|_{i+1/2}$ is the diffusive flux at the interface in electrode j

This automatically conserves lithium in each volume which is FVM's main advantage. For more details on FDM vs. FVM, see [3].

Background: State Estimation with Kalman Filtering

You can't measure lithium concentration, and therefore SOC, directly in a real battery. This is why every BMS needs a state estimator [5, 6].

State-Space Model Development:

1. **Continuous-time model:** The SPM has the form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad y = h(\mathbf{x}, u)$$

where:

- $\mathbf{x} = [\mathbf{c}_{s,n}, \mathbf{c}_{s,p}]^T$ contains concentrations
- $u = I(t)$ is the applied current
- $y = V(t)$ is the terminal voltage

2. **Discrete-time model:** Convert using sampling period Δt :

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d u_k + \mathbf{w}_k, \quad y_k = h(\mathbf{x}_k, u_k) + v_k$$

where:

- $\mathbf{A}_d = e^{\mathbf{A}\Delta t}$ (matrix exponential) or forward Euler: $\mathbf{A}_d = \mathbf{I} + \mathbf{A}\Delta t$
- $\mathbf{B}_d = \mathbf{A}^{-1}(\mathbf{A}_d - \mathbf{I})\mathbf{B}$ or forward Euler: $\mathbf{B}_d = \mathbf{B}\Delta t$
- $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is process noise
- $v_k \sim \mathcal{N}(0, R)$ is measurement noise

3. **Output equation:** The terminal voltage is nonlinear:

$$V(t) = U_p(\theta_p) - U_n(\theta_n) - \eta_p + \eta_n - R_0 I(t)$$

where:

- $\theta = c_{s,\text{surf}}/c_{s,\text{max}}$ is the surface stoichiometry
- U_p, U_n are OCV functions (`U_p.m`, `U_n.m`)
- η_p, η_n are overpotentials (`eta_cathode.m`, `eta_anode.m`)
- R_0 is the lumped resistance (`param.R0`)

Kalman Filter Approaches:

As demonstrated in class lectures and recitations for Equivalent Circuit Models (ECMs), select one of the following special approaches to handle the nonlinear measurement equation:

Extended Kalman Filter (EKF) [6]:

- Linearize the measurement equation using Jacobians: $\mathbf{C}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k}$
- Requires derivatives of OCV and overpotential functions
- Faster but only works well if nonlinearities aren't too strong

Unscented Kalman Filter (UKF):

- Use sigma points instead of Jacobians—no need to compute derivatives
- Generate $2n + 1$ sigma points and propagate them through the model
- Handles strong nonlinearities better but costs more computationally

Adaptive Kalman Filter (AKF) [5]:

- Standard KF/EKF/UKF assume fixed $\mathbf{Q}$ and R —poor choices degrade estimation
- AKF tunes noise covariances online using the innovation (measurement residual) sequence:
 $\epsilon_k = y_k - h(\hat{\mathbf{x}}_{k|k-1}, u_k)$
- A common approach uses a sliding window of length M to estimate R and/or $\mathbf{Q}$:

$$\hat{R}_k = \frac{1}{M} \sum_{i=k-M+1}^k \epsilon_i \epsilon_i^T - \mathbf{C}_k \mathbf{P}_{k|k-1} \mathbf{C}_k^T$$

- Can be combined with either EKF or UKF—yields Adaptive EKF (AEKF) or Adaptive UKF (AUKF)
- Improves robustness when true noise statistics are unknown or time-varying

Kalman Filter Algorithm:

1. **Initialization:** Set $\hat{\mathbf{x}}_0, \mathbf{P}_0, \mathbf{Q}, R$

2. **Prediction Step** (at each time k):

$$\begin{aligned} \hat{\mathbf{x}}_{k|k-1} &= \mathbf{A}_d \hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}_d u_{k-1} \\ \mathbf{P}_{k|k-1} &= \mathbf{A}_d \mathbf{P}_{k-1|k-1} \mathbf{A}_d^T + \mathbf{Q} \end{aligned}$$

3. **Update Step** (when measurement y_k arrives):

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_{k|k-1} \mathbf{C}_k^T (\mathbf{C}_k \mathbf{P}_{k|k-1} \mathbf{C}_k^T + R)^{-1} \\ \hat{\mathbf{x}}_{k|k} &= \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (y_k - h(\hat{\mathbf{x}}_{k|k-1}, u_k)) \\ \mathbf{P}_{k|k} &= (\mathbf{I} - \mathbf{K}_k \mathbf{C}_k) \mathbf{P}_{k|k-1} \end{aligned}$$

4. **Extract SOC:** Compute from average concentration:

$$\text{SOC} = \frac{\theta_j - \theta_j^0}{\theta_j^{100} - \theta_j^0}$$

Where θ_j^0 and θ_j^{100} are the stoichiometries at 0% and 100% SOC, respectively, and θ_j is computed from the estimated surface concentration.

Deliverables

Regardless of which project direction you choose, you must submit:

- **Code Submission:**
 - All MATLAB code developed for your project with comments
 - A README explaining how to run your code
 - Test scripts that reproduce your results
- **Written Report** (max. 12 pages):
(Make sure the first few sections are developed thoroughly from the midterm project report)
 - **Introduction:** Motivation and problem statement
 - **Background:** Brief review of relevant theory
 - **Methodology:** Your approach and implementation details
 - **Results:** Figures, tables, and quantitative analysis
 - **Discussion:** Interpretation of results, limitations, and insights
 - **Conclusions:** Summary and potential future work
- **Presentation** (12 minutes: 10 minutes + 2 minutes Q&A):
 - Clear explanation of your chosen project direction
 - Key results and visualizations
 - Insights and conclusions
 - Q&A with class

Grading Rubric (50 points total)

- **Technical Implementation (10 points):**
 - Correctness of implementation (10 pts)
 - Code quality, organization, and documentation (5 pts)
 - Appropriate use of provided resources and validation (5 pts)
- **Analysis and Results (15 points):**
 - Depth and rigor of analysis (8 pts)
 - Quality of visualizations and quantitative results (4 pts)
 - Insightful interpretation and discussion (3 pts)
- **Written Report (max. 12 pages) (10 points):**
 - Clarity and organization (4 pts)

- Technical writing quality (3 pts)
- Completeness and appropriate scope (3 pts)
- **Presentation (15 points):**
 - Clear communication of methods and results (3 pts)
 - Quality of slides and delivery (2 pts)
 - Q&A performance (10 pts)

References

- [1] Chen, C.-H., Planella, F. B., O'Regan, K., Gastol, D., Widanage, W. D., & Kendrick, E. (2020). Development of experimental techniques for parameterization of multi-scale lithium-ion battery models. *Journal of The Electrochemical Society*, 167(8), 080534.
- [2] Pozzato, G., Allam, A., & Onori, S. (2022). Lithium-ion battery aging dataset based on electric vehicle real-driving profiles. *Data in Brief*, 41, 107995.
- [3] Xu, L., Cooper, J., Allam, A., & Onori, S. (2023). Comparative analysis of numerical methods for lithium-ion battery electrochemical modeling. *Journal of The Electrochemical Society*, 170(12), 120525.
- [4] Lucero, J. N., & Onori, S. (2025, July). Comparative Nonlinear Observability Analysis of Spatial Discretization Schemes for Lithium-Ion Battery Models. *2025 American Control Conference (ACC)*, (pp. 2805-2810), IEEE.
- [5] Taborelli, C., & Onori, S. (2014). Advanced battery management system design for SOC/SOH estimation for e-bikes applications. *International Journal of Powertrains*, 3(4), 382–399.
- [6] Taborelli, C., & Onori, S. (2014). State of charge estimation using extended Kalman filters for battery management system. *2014 IEEE International Electric Vehicle Conference (IEVC)*, (pp. 1–8), IEEE.